

LAB ASSIGNMENT

Lab Assignment 1

The Web and the Browser — hands-on practice with DNS, HTTP, and DevTools



Due: Next Class



4 Tasks



Individual Work



OVERVIEW

What You'll Practice

This lab turns today's lecture into hands-on skills. Complete all four tasks below and submit your work before the next class.

1

DNS report

Practice resolving domains from the terminal

2

curl scavenger hunt

Trigger different HTTP status codes

3

Build & inspect

Extend the Node.js server from class

4

Short written answer

Analyze real network requests



TASKS

Tasks

TASK 1

DNS Report

Run dig (or nslookup) on 3 domains of your choice.

Record for each domain:

- A record (IP address)
- TTL (time to live)
- Authoritative nameservers

Then note how long each TTL is, and guess why.

TASK 2

curl Scavenger Hunt

Find one URL that returns each status code:

200 · 301 or 302 · 404 · 500 or 503

Submit:

- The curl command you used for each
- The status code you got back



TASKS

Tasks

TASK 3

Build & Inspect

Write the 10-line Node.js server from class.

- Add a second route
- Submit a screenshot of its request/response in the DevTools Network tab

TASK 4

Short Written Answer (1/2 page)

Pick one real website.

- Open DevTools → Network
- Describe 3 different requests it made beyond the main HTML document
- Explain what each one was for



LOGISTICS

Submission Guidelines

1

Format

One document (PDF or a shared doc link) with all 4 parts clearly labeled

2

Deadline

Before the start of next class

3

Screenshots

Paste directly into the document, not as separate file attachments

4

Grading

Each of the 4 tasks contributes equally to your grade



PRACTICAL

Tips for Success



Start with Task 1

dig/nslookup issues are easiest to debug early



Keep your terminal history

copy-paste is faster and less error-prone than retyping



No domains handy?

use httpbin.org for Task 2's status codes



Stuck on something?

ask on the class forum — don't wait until the deadline



Good luck!

Questions before next class? Ask on the course forum.